Noticing in Neurology

A J Lees

ABSTRACT

There are three classes of people: those who see, those who see when they are shown, those who do not see

Leonardo da Vinci

The three cardinal qualities necessary for the ideal neurologist are observation, the ability to reason backwards inferentially and specialist knowledge. Modern medical technology has greatly increased the ability to diagnose and treat disease but it has also encouraged a benign variant of abulia, which is killing off the art and science of clinical reasoning. Intent gazing at the unfamiliar with old eyes or a long look at the familiar with new eyes offers the neurologist an opportunity to discover hitherto unnoticed diagnostic signs far beyond the resolution of the brain scanner and even the light microscope. While there may be nothing new under the sun, there are plenty of old things that no one has observed, which have the potential to greatly improve clinical practice.

Correspondence to
Professor A J Lees, Reta Lila Weston Institute of Neurological Studies, University College London, London WC1H 9JR, UK; andrew.lees@ucl.ac.uk

Accepted 28 January 2019

Check for updates

© Author(s) (or their employer(s)) 2019. No commercial re-use. See rights and permissions. Published by BMJ.

To cite: Lees AJ. *Pract Neurol* Epub ahead of print: [please include Day Month Year]. doi:10.1136/practneurol-2018-002176

INTRODUCTION

In his editorial with the tongue in cheek title 'I've stopped examining patients!' (*Practical Neurology* 2009;8:192–4) Christopher Hawkes wrote:

> Increasingly, I find myself examining patients with their clothes on and while they are sitting in the chair, and even more alarmingly (perhaps) I am not examining them at all! Well yes, I do in a sense, as all clinicians do indirectly, by watching patients as they enter the room, by the way they sit, talk, give the history and by observing any involuntary movement.

Acquiring a neurological gaze is an intensely personal process that leads to a distortion in the way the world is viewed. It is an impressionistic myopia that focuses a torchlight on the signs of neurological malfunction but which leaves the rest of the world in darkness. I became competent at detecting slow and hypometric saccades only after John Leigh showed me 'how to see' at an American Academy of Neurology meeting in Chicago. Through the scrutiny of countless patients, a neurologist slowly learns where and how to look. Trying to learn to see nervous disease from photographs in a textbook is like attempting to remember a living flower from a picture.

The neurologist is a private eye looking for clues that will reveal the villain of the piece and as with police work pattern recognition can sometimes solve the crime and provide an instant suspect. But like Sherlock Holmes, neurologists should not transmit their hunches to the patient until all the evidence is in, including information from the clinical history and the focused examination. Necessary qualities needed for diagnostic precision are a more deliberate method of thinking that combines the details of the presenting case with existing knowledge based on previous similar cases, and a capacity to use inference and imagination to reason backwards.

THE ADVENTURE OF THE CARDBOARD BOX

Dr William Gooddy taught me that I would become a neurologist by spending time listening to and examining patients in the hospital, that textbooks were of limited use except as a source of reference and that I should read the Sherlock Holmes canon. In the second of the 12 memoirs of Sherlock Holmes published in The Strand Magazine of 1893, Miss Susan Cushing receives a cardboard box in the post that contains two severed ears packed in salt. Three medical students whom Miss Cushing had been forced to evict for rowdy behaviour had come under suspicion by Inspector Lestrade of Scotland Yard. After making a number of pertinent observations relating to the writing on the box and the knot used to tie up the parcel Holmes turns his attention to the ears and notices that they possess the same broad curve of the upper lobe, the short pinna and convolution of the inner cartilage as Miss Cushing's own ear. At this point Holmes reminds Doctor Watson that

Pract Neurol: first published as 10.1136/practneurol-2018-002176 on 4 April 2019. Downloaded from http://pn.bmj.com/ on 4 April 2019 by guest. Protected by copyright.



Figure 1 The characteristic auricular appearance of DiGeorge syndrome showing a thick overfolded helix and slightly low set ears.

there is no part of the human body that varies as much as the human ear and that he had written two essays on the subject in The Anthropological Journal.

One of Conan Doyle's possible influences for the scientific method of Sherlock Holmes was the work of Giovanni Morelli, an art historian who had studied medicine in Switzerland and Germany, and had taught anatomy at the University of Munich. By paying attention to trifling details such as discrepancies in the portrayal of the human ear, Morelli had acquired a connoisseurship that allowed him to detect several art fakes.

Eight years ago, one of my research fellows commented during a ward round on the low set ears of a young man with Parkinson's syndrome admitted for apomorphine pump therapy. After reminding ourselves of the detailed anatomy of the external ear and consulting an atlas of dysmorphic faces, we concluded that the patient had a thick overfolded helix, slightly cup-shaped ears and deep pits (figure 1). This led to the confirmation of a microdeletion in the 22q11.2 chromosome (DiGeorge syndrome). A subsequent literature search revealed that several patients with young onset L-dopa-responsive Parkinson's syndrome had recently been reported with the same chromosome abnormality.

SIGNS ON THE LONDON UNDERGROUND

When I was a junior registrar at University College Hospital, I used to take the medical students on the London Underground Circle Line in order to hone their powers of observation. Before leaving the hospital they were instructed to pay special attention to gaits and facial expression. No cameras were allowed and I warned them not to ogle. On the westbound platform at Euston Square, we would split into pairs and enter different compartments. On return to the day room of Ward 5.2 just over an hour later, they were asked to provide me with statements in much the same way they would do to the London Transport police if they had witnessed a crime. A hang dog appearance, abnormal shouting, absent arm swing, a grand mal fit and a sailor's walk were some of the descriptions I can remember that we discussed. There was no emphasis on semiotics or attempts to guess the diagnosis. It was an exercise in sentimental empiricism, field work of the sort Walter Benjamin called 'botanizing on the asphalt'[1] that I hoped would teach them the difference between observation and inference. It helped them appreciate that an image is not created in the retina but is constructed in the brain and given shape through verbal or written description.

On other days, I would take the students on the ward and after introducing them to a patient ask them to look for a few moments. After promising to return at the end of the teaching session, we would troop out and return to the day room. Over time they became able to see more than a sick person in a bed and look beyond the intravenous drip and the hospital paraphernalia and begin to imagine a picture that embraced the patient's whole life, her home, work, hopes and sorrows as well as her disease.

In *A Study in Scarlet*, Doctor Watson came across an article in a magazine entitled *The Book of Life* that unbeknown to him had been written anonymously by Sherlock Holmes, in which it was claimed that a man's innermost thoughts could be divined by a momentary expression, a twitch of a muscle or a glance of an eye. The author had gone on to argue that from a drop of water, a logician could infer the possibility of an Atlantic or a Niagara without ever having seen or heard of one or the other. When Watson cried out in the living room at 221B Baker Street that the piece was 'ineffable twaddle' and that he would like to bet that if the writer were clapped down in a third-class carriage on the London Underground, there would be less than a thousand to one chance he could name the trades of his fellow passengers, Holmes smiled and told him that he would certainly lose his money.

After I had become a consultant, I found it hard to leave my work behind and started to pay attention to physiognomy on the London Underground.[2 3] The fraught black line with its anxious commuters offered up a surreptitious naturalistic observation chamber where unlike the doctor's surgery people did not 'put on a face' (figure 2). The notebook where I had recorded these observations between 1981 and 1984 miraculously came to light in a long-forgotten box of

Pract Neurol: first published as 10.1136/practneurol-2018-002176 on 4 April 2019. Downloaded from http://pn.bmj.com/ on 4 April 2019 by guest. Protected by copyright.



Figure 2 Subway neurology.

junk in my office and was the prompt for this article. Table 1 lists of some of the abnormal movements that I have never encountered in the clinic and my retrospective guesses of what might have been responsible.

Spot diagnoses in neurology are possible but run the risk of jumping to an erroneous conclusion or overvaluing what is seen in the flawed belief that the 'internal site' of a disease can be diagnosed with absolute precision from its 'specific external signs'. Because inspection is such a personal matter, neurologists frequently report findings that no two other eyes can see, running the risk of an outbreak of blind obedience (Emperor's New Clothes syndrome) among their juniors. The 'glance of the eye' diagnosis, once popular in the short cases of the Membership of the Royal College of Physicians examination has now rightly disappeared and been replaced by the practical assessment of clinical examination skills. What I was doing on the Underground was not diagnosing but learning to notice.

THE GRAND MASTERS

To Jean-Martin Charcot symptoms and signs were the external manifestations of a distressed brain; like a Victorian botanist, his greatest satisfaction was to describe in minute detail something commonplace, never previously recognised. Before his arrival each morning at the Salpêtrière Hospital, his interns were expected to have taken a detailed clinical history and conducted the examination (together referred to as *L'Observation*). Charcot would stare intently at the patient as she was escorted into the small room that had once served as a kitchen and without averting his gaze listen to the history read out by his assistants. After what often seemed like an interminable pause, he might ask the patient to make a movement or speak or turn to his intern and ask for some further detail on the family history. His *chef de clinique* might then be asked to elicit a physical sign. Sometimes, a second and third patient was brought in from the wards for comparison.

In his Tuesday lessons Charcot would sometimes ask rhetorically why it was that most doctors could only see the things they have been taught and expect to encounter. On 12 June 1888, he taught on the shaking palsy and after commending James Parkinson's Essay to his students, he went on to say that he had seen patients with the same picture on the streets of Rome, of Amsterdam and Spain and that they could be identified from afar without the need for a medical history. Sitting on a dais above his audience, he then fixed the patient with his *coup d'oeuil* and after a pause described what he could see (figure 3):

> *The muscles of the face are motionless there is even a remarkable fixity of look, and the features present a permanent expression of mournfulness, sometimes of stolidness or stupidity… This particular face was*

Table 1 Some of the Abnormal movements I observed on the Northern Line train between East Finchley and Warren Street in the early eighties

Commuter	Location	Facial movements	Other observations	Possible cause
Thin nervy young man	Camden Town	Continuous stereotyped jaw movements and lip pursing	Scar on lip, picked nails, acne, dilated pupils	Chronic amphetamine use
Young woman with furrowed brow	Tufnell Park	Persistent left lower lip pulling down in a smirk		Functional movement disorder
A middle-aged man with glasses in pin-striped suit	Highgate	Fixed stare with hardly a blink over 30 min observation	Unnatural stillness of upper body	Early Parkinson's disease
A weary woman in her 40s who kept dozing	Mornington Crescent	Quivering of right lower eyelid with fine undulating non-synchronous contractions occurring several times each second lasting 30 s	Placing of a finger below the eye where twitching was occurring	Eyelid myokymia
Woman in her 30s	Camden Town	Complex persistent side to side and vertical larynx movements	Repetitive foot tapping	'Jumping larynx' possibly linked to palatal tremor? or severe anxiety
A boy with his mother	Warren Street	Left eyelid partially closed on yawning on two occasions.	Normal eye movements and no facial weakness	Marcus Gunn phenomenon
Dishevelled obese man with badly fitting clothes	Kentish Town	Continuous slow tremor of mouth and lip	Nicotine-stained hands	Rabbit syndrome due to neuroleptics

Pract Neurol: first published as 10.1136/practneurol-2018-002176 on 4 April 2019. Downloaded from http://pn.bmj.com/ on 4 April 2019 by guest. Protected by copyright.



Figure 3 A drawing by Paul Richer of Charcot delivering a Leçon du Mardi at La Salpêtrière. École Nationale Supérieure des Beaux-Arts (Paris), with kind permission.

> *not originally appreciated. It is not in Parkinson's description*.

Charcot had toyed with becoming a painter before entering medical school and for him drawing and illustration represented a branch of science. He encouraged his assistants to observe the works of the Grand Masters in the Louvre for examples of nervous maladies, and in 1888 launched *La Nouvelle Iconographie de la Salpêtrière* with colour lithographs, photographs and drawings of clinical cases.

To his students it seemed at times that he had conjured a new clinical syndrome out of a hat but in his lectures he always emphasised that his method involved repeated observation carried out over an extended period of time. Patterns and connections appeared in the phenomena that marched before his eyes and over time chaos gave way to order and a new knowledge. At a dinner party at his home he told Sigmund Freud '*je n'suis qu'un visuel*'.

William Gowers was also a skilled artist who embellished his word pictures with pen and ink drawings. In the *Manual of the Diseases of the Nervous System*, he emphasised the importance of precise observation and the absolute difference between looking and seeing. Much of his reputation as the consummate diagnostician stemmed from his ability to see shared similarities and resemblances in seemingly disparate cases. In a clinical demonstration at the National Hospital Queen Square, he reminded the audience of the usefulness of inspection:

> Did you notice him as he came into the room? If you did not then you should have done so. One of the habits to be acquired and never omitted is to observe a patient as he enters the room; to note his aspect and his gait. If you did so, you would have seen that he seemed lame, and you may have been struck by that which must strike you now—an unusual tint of his face'

HOW TO BECOME A NOTICER

Noticing takes time and patience to acquire. To visualise afresh what on eyeballing appears to be mundane and see 'The Invisible Gorilla' -the whole picture and not a part[4] - often involves a return to the panoramic visual curiosity of childhood. With constant practice, the eye can be trained to see what the mind does not know. To become a seer requires the brow to be raised, the blink suppressed and the patient to be held in view as still as if a photograph was being taken. Keeping the eyes wide open involves perceiving what is absent as well as what is present.

The diagnostic criteria and rating scales increasingly used in neurology assume a competence of observation that is lacking in some of those it is hoped will use them. A rich knowledge base and an ability to reflect and analyse is of little use if one has never been taught through a deficit in training how to reliably acquire raw data.

The more technology that neurologists have at their disposal and the more complex they are, the greater the demands will become on their clinical judgement. An overdependence on sophisticated gadgetry deters neurologists from using their eyes and their brain. Intuitive inspection will always be more informative and revealing than any single finding from a machine.

Acknowledgements Drs Olivier Wasulinski and Christopher Goetz for details relating to Charcot's clinical method.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Not required.

Provenance and peer review Commissioned. Externally peer reviewed by Kevin Talbot, Oxford, UK.

REFERENCES

1 Benjamin W, Eiland H, MacLaughlin K. *The arcades project*. Cambridge: Belknap Press, 1999.
2 Lees AJ. Movement disorders in psychiatry. *Curr Opin Neurol Neurosurg* 1990;3:342–5.
3 Chabris CF, Simons DJ. *The invisible gorilla : and other ways our intuitions deceive us*. New York: Crown, 2010.
4 Lees AJ. Facial mannerisms and tics. *Adv Neurol* 1988;49:255–61.

Pract Neurol: first published as 10.1136/practneurol-2018-002176 on 4 April 2019. Downloaded from http://pn.bmj.com/ on 4 April 2019 by guest. Protected by copyright.